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Comparative performance of common paediatric patient-reported outcome measures (P-PROMs) across health conditions

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ABSTRACT

Objectives To compare the measurement properties of paediatric patient-reported outcome measures (P-PROMs) used in clinical trials and practice for common conditions.

Design Data from the Australian Paediatric Multi-Instrument Comparison study were used, including children aged 5–18 years with autism spectrum disorder (ASD) (n=510), asthma (n=487), eating disorder(s) (n=216), epilepsy (n=298), recurrent abdominal pain (n=392), sleep problems (n=346), dental problems (n=490), type 1 diabetes (T1D) (n=67), enuresis (n=182) and no health conditions (reference group) (n=1259). The acceptability, response distribution (floor/ceiling effects), known-groups validity, test-retest reliability, responsiveness and convergent validity of the Paediatric Quality of Life Inventory (PedsQL), EuroQol Group's Youth instruments with three and five response options (EQ-5D-Y-3L and EQ-5D-Y-5L), Child Health Utility 9D (CHU9D), Assessment of Quality of Life 6D (AQoL-6D), Health Utilities Index Mark 3 and Patient-Reported Outcomes Measurement Information System Paediatric Profile 25 (PROMIS-25) were assessed.

Results PedsQL, EQ-5D-Y-3L, EQ-5D-Y-5L, CHU9D and PROMIS-25 were easy and quick to complete, indicating acceptability. Only the EQ-5D-Y-3L and EQ-5D-Y-5L demonstrated ceiling effect issues. All measures were able to differentiate between children with no health conditions and children with each condition ($p < 0.001$, effect sizes between 0.47 and 3.5), indicating known-groups validity. Test-retest reliability varied by condition; however, the PedsQL and EQ-5D-Y-5L were the most reliable.

Conclusions Certain P-PROMs may be preferred in different conditions, that is, ASD (PedsQL and EQ-5D-Y-5L), asthma (PedsQL), eating disorder(s) (PedsQL and CHU9D), epilepsy (PedsQL and CHU9D), recurrent abdominal pain (PedsQL, EQ-5D-Y-5L and CHU9D), sleep problems (PedsQL), dental problems (PedsQL), T1D (PedsQL and CHU9D) and enuresis (PedsQL). Notably, the PedsQL performed favourably in all conditions assessed. More evidence is required on the responsiveness of P-PROMs to changes in child health.

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INTRODUCTION

Generic paediatric patient-reported outcome measures (P-PROMs) capture a child's health-related

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Paediatric patient-reported outcome measures are increasingly used in clinical care and research to provide information on a child's overall health. However, there is limited research comparing the psychometric performance of these measures in different conditions, making measurement selection somewhat arbitrary.

WHAT THIS STUDY ADDS

⇒ This study is the first globally to directly compare the performance of commonly used measures of overall paediatric health across a range of common conditions.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The findings provide researchers and clinicians with the evidence to make informed choices when selecting a paediatric patient-reported outcome measure for use in clinical trials, research or clinical care.

quality of life (HRQoL), including physical, social and emotional well-being.¹ Common generic P-PROMs include the Paediatric Quality of Life Inventory (PedsQL), EuroQol Group's Youth instruments with three and five response options (EQ-5D-Y-3L and EQ-5D-Y-5L), Child Health Utility 9D (CHU9D), Assessment of Quality of Life 6D (AQoL-6D), Health Utilities Index Mark 2 and 3 (HUI2 and HUI3) and Patient-Reported Outcomes Measurement Information System Paediatric Profile 25 (PROMIS-25).^{2,3} Unlike condition-specific P-PROMs, generic P-PROMs capture aspects of health that are common to most children, meaning they can be used across populations and conditions.⁴ Generic P-PROMs have historically been used in clinical trials, studies and registries^{1,4,5} but are increasingly being used in routine clinical care to enable more patient-centred care.⁶ Given their breadth of use, it is crucial that these measures perform well in the populations they are used in.

There is a lack of evidence regarding the comparative performance of generic P-PROMs, prohibiting future users from making an informed choice as



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to which measure may be best for their intended use or population.^{2 3 7} Recent systematic reviews specifically highlight the limited evidence on the test-retest reliability and responsiveness of generic P-PROMs.^{2 3} When used in a clinical study or to inform practice, a measure's sensitivity to detect changes in a child's health is especially important. Additionally, evidence about the performance of measures in certain conditions is often limited by small samples or studies only having collected a single measure, precluding head-to-head comparisons.^{2 3} Other research has attempted to generate a standard set of health outcomes for all children using consensus-based methods, including both non-patient-reported outcomes and generic P-PROMs.⁸ The generic P-PROMs recommended from this consensus research included PROMIS and AQoL measures.⁸ However, given the lack of comparative evidence on the performance of generic P-PROMs at the time of the consensus study, decision-making is largely based on expert opinion and face validity. Hence, further comparative evidence on the measurement performance of common generic P-PROMs in different condition groups is required.

The Australian Paediatric Multi-Instrument Comparison (P-MIC) study is the largest and most comprehensive comparative study of P-PROMs in the world. The study has published evidence on the comparative performance of generic P-PROMs in samples of children with and without health conditions and in children with mental health challenges.^{9 10} However, no previous publications have assessed the performance of the full range of generic P-PROMs collected in the P-MIC study, nor how these P-PROMs perform for children with clinical conditions. Existing evidence highlights that different generic P-PROMs perform better in different conditions,^{9 10} further emphasising the need to assess the comparative performance of measures in as many conditions as possible. Using survey data from the Australian P-MIC study, this paper aims to compare the measurement properties of common generic P-PROMs (PedsQL, EQ-5D-Y-3L, EQ-5D-Y-5L, CHU9D, AQoL-6D, HUI3 and PROMIS-25) in samples from the P-MIC study reporting common conditions, outlined below. This information will be helpful to researchers and clinicians seeking to make selections about P-PROMs for inclusion in their studies and clinical practice.

METHODS

Data were from the Australian P-MIC study (ethics approval HREC/71872/RCHM2021), which collected a range of P-PROMs head-to-head at two time points.^{11 12} Specific condition samples were included in the P-MIC to enable measurement assessment in condition groups. These conditions were selected based on set criteria, including evidence of poorer HRQoL and P-PROMs not yet having substantial evidence of validity in the condition.¹²

Participants

The following nine P-MIC condition samples were included in analysis: autism spectrum disorder (ASD), asthma, eating disorders (including anorexia, bulimia and avoidant restrictive food intake disorder), epilepsy, recurrent abdominal pain, sleep problems, dental problems (including self-reported caries experience, malocclusion and dental trauma), type 1 diabetes (T1D) and enuresis. To determine eligibility for condition samples, caregivers were asked to complete screening questions (online supplemental table 1). A hierarchical recruitment approach was used, whereby caregivers were asked to select which condition samples their child/children were eligible for. If a child were eligible for multiple conditions, they were recruited into the

rarest condition sample.¹² If they were not eligible for any, they were recruited into the general population sample (healthy reference group).¹²

All children in this analysis were aged 5–18 years; younger children were excluded, given the experimental nature of measures in this age group.^{11 12} Replicating methods from prior work, the healthy reference group included participants from the general population sample whose caregiver reported no health conditions and ≥ 70 on the EuroQoL Visual Analogue Scale (EQ VAS, indicating good overall health).¹²

P-MIC participants were recruited from a community setting using largely an online panel company, Pureprofile Australia. Some participants from the epilepsy (32%) and eating disorder (20%) samples were recruited via online supplemental methods managed by the study team (eg, advertising through hospital telehealth appointments or in-person clinics, social media and community newsletters) to top up online panel recruitment.

Data collection

Eligible participants consented and completed an initial survey online between November 2021 and June 2023. All participants included in this analysis were asked to complete a follow-up survey 4 weeks later.

In the initial survey, participants (caregivers) completed demographic questions. Participants (child or caregiver) then completed the PedsQL, EQ-5D-Y-3L, EQ-5D-Y-5L and CHU9D (order randomised to minimise order effects). To minimise responder burden, participants (child or caregiver) were then randomly allocated to complete either the AQoL-6D, HUI2/3 or PROMIS-25. Finally, participants (child or caregiver) completed a condition-specific measure that corresponded to their condition group (online supplemental table 2), enabling assessment of condition severity and symptoms used for known-groups validity assessment.¹² No suitable ASD measure was identified; hence the Strengths and Difficulties Questionnaire, a behavioural well-being measure, was used.¹³ Condition-specific measures were not included in the follow-up survey to minimise responder burden. Measure characteristics are summarised in online supplemental table 2.

Children were asked to self-report P-PROMs if they were 7 years or older and were able to answer questions about their health. If the child was unable to self-report, the caregiver was asked to proxy report. The condition-specific measures in the epilepsy and sleep problems samples were always proxy-reported, as only proxy report versions were available.

Scoring measures

To ensure comparability of all measures with previous studies,^{9 10} a total score for the EQ-5D-Y-3L, EQ-5D-Y-5L, CHU9D, AQoL-6D and HUI3 was calculated using a level sum score approach.^{14 15} A total score for the PedsQL was calculated as per developer instructions.¹⁶ Although the PROMIS-25 is designed to be scored at the domain level, many users require a single total summary score of some type. This is recognised by PROMIS developers who have generated a value set to enable the adult version, the PROMIS-29, to be transformed into a single utility score.¹⁷ However, no such value set exists for the PROMIS-25. Consequently, an experimental scoring approach was used in this study for the PROMIS-25 to generate a single total score to enable direct comparison to other measures.¹⁸ This involved taking the mean T-score of all six domains, inverting the T-score on the anxiety, depressive symptoms, fatigue and pain domains to ensure all T-scores were in the same direction (ie,

Table 1 Participant characteristics

Participant characteristic		Sample									
		Autism spectrum disorder	Asthma	Eating disorder	Epilepsy	Recurrent abdominal pain	Sleep problems	Dental problems	Type 1 diabetes	Enuresis	Healthy reference group
Completed initial survey, n (%)		510 (100)	487 (100)	216 (100)	298 (100)	392 (100)	346 (100)	490 (100)	67 (100)	182 (100)	1259 (100)
Completed follow-up survey, n (%)		175 (34.3)	173 (35.5)	95 (44.0)	138 (46.3)	127 (32.4)	96 (27.8)	144 (29.4)	30 (44.8)	91 (50)	493 (39.2)
Completed core HRQoL instruments (CHU9D, PedsQL, EQ-5D-Y-3L and EQ-5D-Y-5L), n (%)		510 (100)	487 (100)	216 (100)	298 (100)	392 (100)	346 (100)	490 (100)	67 (100)	182 (100)	1259 (100)
Completed AQoL-6D, n (%)		66 (12.9)	172 (35.3)	60 (27.8)	99 (33.2)	40 (10.2)	100 (28.9)	174 (35.5)	24 (35.8)	58 (31.9)	405 (32.2)
Completed HUI 2/3, n (%)		153 (30.0)	155 (31.8)	80 (37.0)	95 (31.9)	133 (33.9)	121 (35.0)	158 (32.2)	24 (35.8)	64 (35.2)	444 (35.3)
Completed PROMIS-25, n (%)		184 (36.1)	160 (32.9)	76 (35.2)	104 (34.9)	121 (30.9)	125 (36.1)	158 (32.2)	19 (28.4)	60 (33)	410 (32.6)
Proxy reported HRQoL instruments*		183 (35.9)	185 (38)	50 (23.2)	108 (36.2)	124 (31.6)	156 (45.1)	173 (35.3)	25 (37.3)	75 (41.2)	453 (36.0)
Child age, range, mean (SD)		5–18 10.6 (3.7)	5–18 9.9 (3.6)	14–18 15.7 (1.3)	5–18 10.9 (3.7)	5–18 10.2 (3.6)	5–16 9.1 (3.5)	5-18 9.4 (3.5)	5-18 10.1 (3.7)	6-17 8.7 (2.7)	5-18 11.1 (3.9)
Child gender, n (%)	Female	140 (27.5)	184 (37.8)	161 (74.5)	161 (54.0)	254 (64.8)	166 (48)	224 (45.7)	25 (37.3)	89 (48.9)	608 (48.3)
	Male	358 (70.2)	301 (61.8)	42 (19.4)	134 (45.0)	135 (34.4)	176 (50.9)	263 (53.7)	42 (62.7)	93 (51.1)	647 (51.4)
	Prefer not to answer/ other	12 (2.4)	2 (0.4)	13 (6.0)	3 (1.0)	3 (0.8)	4 (1.2)	3 (0.6)	0 (0.0)	0 (0.0)	4 (0.3)
Child of Aboriginal and/or Torres Strait Islander origin—yes, n (%)		46 (9.0)	46 (9.5)	23 (10.6)	21 (7.1)	23 (5.9)	26 (7.5)	32 (6.5)	6 (9.0)	13 (7.1)	40 (3.2)
Child speaks language other than English spoken at home—yes, n (%)		23 (4.5)	32 (6.6)	18 (8.3)	17 (5.7)	18 (4.6)	14 (4.1)	66 (13.5)	8 (11.9)	7 (3.9)	170 (13.5)
Caregiver age, mean (SD)		40.1 (8.5)	39.0 (8.1)	43.4 (9.6)	39.8 (7.6)	42.4 (9.1)	39.1 (7.5)	38.2 (7.5)	38.7 (8.0)	40.6 (9)	38.9 (7.5)
Caregiver highest education level—bachelor degree or above, n (%)		134 (26.3)	170 (34.9)	79 (36.6)	128 (43)	123 (31.4)	101 (29.2)	206 (42)	37 (55.2)	65 (35.7)	518 (41.1)
*Counterfactual to proxy report is child self-report.											
AQoL-6D, Assessment of Quality of Life; CHU9D, Child Health Utility; EQ-5D-Y-3L, EuroQol Group's Youth instruments with three response; EQ-5D-Y-5L, EuroQol Group's Youth instruments with five response; HRQoL, health-related quality of life; HUI3, Health Utilities Index Mark 3; PedsQL, Paediatric Quality of Life Inventory.											

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AQoL-6D, Assessment of Quality of Life; CHU9D, Child Health Utility; EQ-5D-Y-3L, EuroQol Group's Youth instruments with three response; EQ-5D-Y-5L, EuroQol Group's Youth instruments with five response; HRQoL, health-related quality of life; HUI3, Health Utilities Index Mark 3; PedsQL, Paediatric Quality of Life Inventory.

higher score equals better health). Where sample sizes were large enough, sensitivity analyses were conducted for each psychometric attribute using the preferred PROMIS scoring approach (T scores for each domain). The direction and range of scores differ for each measure and are described in online supplemental table 2 alongside details regarding condition-specific measures.

Statistical analysis

Statistical methods used to evaluate measurement performance are described in online supplemental table 3.^{19–28} These methods, including hypotheses and thresholds for significance, are also described in the P-MIC methods guide.¹² If sample sizes were too small (ie, 'inadequate' or 'doubtful' as per Consensus-based Standards for the selection of health Measurement INstruments (COSMIN) guidelines),²⁷ results were not interpreted. All analyses were completed in Stata V.17 (StataCorp, 2022, Release 17).

RESULTS

Participant characteristics are described in table 1. Participants included children aged 5–18 years with ASD (n=510), asthma (n=487), eating disorder(s) (n=216), epilepsy (n=298), recurrent abdominal pain (n=392), sleep problems (n=346), dental problems (n=490), T1D (n=67), enuresis (n=182) and no health conditions (reference group) (n=1259). 4-week follow-up survey completion rates varied by condition, ranging from 28% (sleep problems) to 50% (enuresis).

Acceptability

Participant self-reported difficulty completing each measure is presented in online supplemental figure 1. There was a significant difference in the participant-reported ease/difficulty completing measures across conditions ($p<0.001$); however, within each

condition, participants generally reported similar ease/difficulty (online supplemental table 4).

The mean time (s) to complete each measure is presented in online supplemental table 5. The EQ-5D-Y-3L (48.0–73.8 s) and EQ-5D-Y-5L (42.3–60.7 s) were generally the quickest, and the AQoL-6D (107.3–202.9 s) and HUI3 (140.9–224.9 s) were generally the longest.

Distribution of responses

The distributions of responses for each measure are presented in online supplemental figures 2–8. The ceiling effects of each measure (proportion of respondents who report the best possible HRQoL on the measure) are presented in online supplemental table 6. The EQ-5D-Y-3L, EQ-5D-Y-5L and PROMIS-25 demonstrated ceiling effect issues (>15% of participants reporting the best possible HRQoL) in the asthma group. The EQ-5D-Y-3L and EQ-5D-Y-5L also demonstrated ceiling effect issues in the epilepsy, sleep problems, dental problems, T1D and enuresis groups. The extent of ceiling effects varied, with the asthma group having the most pronounced ceiling effects (EQ-5D-Y-3L: 50.9% and EQ-5D-Y-5L: 41.7%), and the sleep problems group having the least pronounced (EQ-5D-Y-3L: 22.3% and EQ-5D-Y-5L: 17.3%). No measure demonstrated floor effects.

Known-groups validity

Known-groups validity (ie, ability to differentiate between groups who were hypothesised to have different HRQoL) results are summarised in online supplemental table 7. Where sample sizes were large enough to interpret results, all measures were able to detect a statistically significant difference ($p<0.05$) between children from the healthy reference group and children from each condition group, with a large effect size, except for

the asthma and dental problems condition groups, where effect sizes were moderate or small.

Furthermore, where sample sizes were large enough to interpret results, all measures were able to detect a statistically significant difference ($p < 0.05$) between those with less and more severe disease (using condition-specific measure cut points). Effect sizes were moderate or large for all comparisons, except the enuresis group, where effect sizes were small for the EQ-5D-Y-3L and EQ-5D-Y-5L.

Test-retest reliability

The test-retest reliability of each measure (the degree to which the measure score remains stable when the participant's health remains stable) is summarised in table 2. In the ASD group, the PedsQL, EQ-5D-Y-3L and EQ-5D-Y-5L demonstrated moderate test-retest reliability, while the CHU9D demonstrated poor test-retest reliability. However, CIs ranged from poor to moderate. In the asthma group, only the EQ-5D-Y-5L demonstrated moderate test-retest reliability, although again, CIs ranged from poor to moderate. In the epilepsy group, the PedsQL, EQ-5D-Y-3L, EQ-5D-Y-5L and CHU9D demonstrated moderate test-retest reliability, with CIs remaining within the moderate range. In the dental problems group, only the PedsQL and CHU9D demonstrated moderate test-retest reliability; however, CIs ranged from poor to moderate. In the enuresis group, the PedsQL, EQ-5D-Y-3L and EQ-5D-Y-5L demonstrated good test-retest reliability, and CHU9D demonstrated moderate test-retest reliability. All CIs remained in the moderate-good range. Sample sizes for the AQoL-6D, HUI3 and PROMIS-25 were too small to interpret results, as were sample sizes for all measures in the eating disorders, recurrent abdominal pain, sleep problems and T1D groups.

Responsiveness

The responsiveness of measures (the ability to detect a caregiver reporting an improved change in a child's main health condition) is summarised in table 3. Sample sizes were only large enough to robustly assess responsiveness for the asthma, recurrent abdominal pain and dental problems groups and only for the PedsQL, EQ-5D-Y-3L, EQ-5D-Y-5L and CHU9D. None of the measures detected a statistically significant difference in total score among caregivers/children who had reported an improvement in health between time points. Sample sizes for the AQoL-6D, HUI3 and PROMIS-25 were too small to robustly interpret responsiveness results, as were sample sizes for all measures in the remaining condition groups.

Convergent validity

The convergent validity of measures (the degree to which measures are correlated where hypothesised) is summarised in online supplemental table 8. Except for the HUI3 in the recurrent abdominal pain sample that was moderately correlated with all other measures, all measures were strongly correlated with each other, indicating convergent validity.

Sensitivity analyses (PROMIS-25 domain T-scores)

Inconsistency between the ceiling effect and known-groups validity results for the PROMIS-25 was identified when scored using the total score versus domain T scores (as per developer instructions; online supplemental tables 10 and 11). For example, when the total score was used, the PROMIS-25 only displayed ceiling effect issues in the asthma sample; however, when the domain scoring was used, at least two of the PROMIS-25 domains displayed ceiling effect issues in all condition groups.

Table 2 Test-retest reliability by condition group

Instruments	ASD		Asthma		Eating disorder		Epilepsy		Recurrent abdominal pain		Sleep problems		Dental problems		Type 1 diabetes		Enuresis	
	N	ICC (95% CI)	N	ICC (95% CI)	N	ICC (95% CI)	N	ICC (95% CI)	N	ICC (95% CI)	N	ICC (95% CI)	N	ICC (95% CI)	N	ICC (95% CI)	N	ICC (95% CI)
PedsQL	106	0.56 (0.39 to 0.69)	93	0.41 (0.23 to 0.57)	32	0.57 (0.27 to 0.76)	77	0.66 (0.51 to 0.77)	47	0.76 (0.60 to 0.86)	38	0.67 (0.44 to 0.81)	58	0.50 (0.28 to 0.67)	14	0.84 (0.59 to 0.95)	51	0.81 (0.68 to 0.88)
EQ-5D-Y-3L	106	0.58 (0.43 to 0.69)	93	0.33 (0.14 to 0.50)	32	0.61 (0.33 to 0.79)	77	0.68 (0.54 to 0.79)	47	0.73 (0.56 to 0.84)	38	0.35 (0.04 to 0.60)	58	0.22 (−0.04 to 0.45)	14	0.29 (−0.30 to 0.71)	51	0.79 (0.66 to 0.88)
EQ-5D-Y-5L	105	0.60 (0.47 to 0.71)	92	0.51 (0.35 to 0.65)	31	0.55 (0.25 to 0.75)	76	0.74 (0.62 to 0.83)	44	0.73 (0.55 to 0.84)	38	0.49 (0.20 to 0.70)	58	0.45 (0.22 to 0.63)	13	0.18 (−0.43 to 0.66)	50	0.80 (0.67 to 0.88)
CHU9D	106	0.42 (0.22 to 0.60)	93	0.48 (0.31 to 0.62)	32	0.53 (0.23 to 0.74)	77	0.64 (0.48 to 0.76)	47	0.66 (0.46 to 0.80)	38	0.70 (0.49 to 0.84)	58	0.50 (0.27 to 0.67)	14	0.34 (−0.23 to 0.73)	51	0.69 (0.51 to 0.81)
AQoL-6D	13	0.51 (0 to 0.81)	26	0.53 (0.20 to 0.76)	6	0.09 (−0.57 to 0.78)	27	0.57 (0.26 to 0.78)	4	0.30 (−0.58 to 0.93)	9	0.80 (0.36 to 0.95)	25	0.24 (−0.18 to 0.58)	6	0.76 (−0.07 to 0.96)	18	0.88 (0.72 to 0.95)
HUI 3	29	0.64 (0.36 to 0.81)	34	0.64 (0.39 to 0.80)	14	0.52 (0.01 to 0.82)	29	0.64 (0.34 to 0.81)	17	0.60 (0.17 to 0.84)	10	0.17 (−0.54 to 0.71)	17	0.77 (0.48 to 0.91)	2	0 (−4 to 1)	17	0.64 (0.25 to 0.85)
PROMIS-25	46	0.44 (0.18 to 0.64)	33	0.35 (0.01 to 0.62)	12	0.73 (0.29 to 0.92)	21	0.87 (0.72 to 0.95)	14	0.76 (0.41 to 0.92)	19	0.66 (0.32 to 0.85)	16	0.28 (−0.26 to 0.68)	6	0.89 (0.49 to 0.98)	16	0.83 (0.59 to 0.94)
Green indicates good or excellent agreement (ICC ≥0.75). Yellow indicates moderate agreement (ICC 0.5–0.74). White indicates poor agreement (ICC <0.5). ³⁵ Results for sample sizes less than 50 are greyed out.																		
AQoL-6D, Assessment of Quality of Life; CHU9D, Child Health Utility; EQ-5D-Y-3L, EuroQol Group's Youth instruments with three response; EQ-5D-Y-5L, EuroQol Group's Youth instruments with five response; HUI3, Health Utilities Index Mark 3; ICC, intraclass correlation coefficient; PedsQL, Paediatric Quality of Life Inventory																		

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AQoL-6D, Assessment of Quality of Life; CHU9D, Child Health Utility; EQ-5D-Y-3L, EuroQol Group's Youth Instruments with three response; EQ-5D-Y-5L, EuroQol Group's Youth Instruments with five response; HUI3, Health Utilities Index Mark 3; ICC, Intraclass correlation coefficient; PedsQL, Paediatric Quality of Life Inventory.

Table 3 Instrument responsiveness, by condition group

Condition group	Instrument	N	N participants reporting improvement in main health condition, mean difference in total score between time points, P value and SRM of mean difference		
			Mean difference, total score time point 1–total score time point 2 (SD)	P value	SRM
Asthma	PedsQL	57	3.62 (17.7)	0.9360	–0.205
	EQ-5D-Y-3L	57	0.04 (1.72)	0.4391	–0.020
	EQ-5D-Y-5L	57	0.11 (2.0)	0.3468	–0.053
	CHU9D	57	0.58 (5.6)	0.2192	–0.103
Recurrent abdominal pain	PedsQL	67	–1.91 (15.6)	0.1595	0.123
	EQ-5D-Y-3L	67	0.36 (1.8)	0.0514	–0.202
	EQ-5D-Y-5L	67	0.20 (2.3)	0.2429	–0.087
	CHU9D	67	1.51 (5.8)	0.0181	–0.261
Dental problems	PedsQL	71	1.78 (17.3)	0.7997	–0.103
	EQ-5D-Y-3L	71	–0.06 (1.3)	0.6433	0.045
	EQ-5D-Y-5L	71	0.08 (1.9)	0.3704	–0.042
	CHU9D	71	0.15 (4.6)	0.3946	–0.033

Green indicates large effect size (SRM ≥ 0.8), yellow indicates moderate effect size (SRM 0.5–0.79), and white indicates small effect size (SRM 0.2–0.49).^{37,38} Bold indicated $p < 0.05$. Results for sample sizes less than 50 are not displayed. Mean differences going in the opposite direction to what is hypothesised are highlighted in orange.

AQoL-6D, Assessment of Quality of Life; CHU9D, Child Health Utility; EQ-5D-Y-3L, EuroQol Group's Youth instruments with three response; EQ-5D-Y-5L, EuroQol Group's Youth instruments with five response; HUI3, Health Utilities Index Mark 3; PedsQL, Paediatric Quality of Life Inventory; SRM, standardised response mean.

Sample sizes were not large enough to enable analysis of test-retest reliability or responsiveness (online supplemental tables 9–11).

Summary of measure performance

Online supplemental table 12 provides a summary of how the measures perform in each condition group across the different measurement attributes assessed.

DISCUSSION

This is the only study to provide a head-to-head comparison of the performance of commonly used P-PROMs across a range of common paediatric conditions.²⁸ The acceptability and convergent validity of measures were similarly good in all condition groups; however, the response distribution, known-groups validity, test-retest reliability and responsiveness of measures varied by condition. Despite these variations, the PedsQL was found to perform well across these measurement attributes for all conditions assessed.

A major strength of this study is the inclusion of a range of condition groups with dedicated sample sizes to enable a head-to-head comparison of measures within condition groups.^{2 3 7} Furthermore, we included conditions for which there had been no or limited evidence. Consequently, this study fills distinct knowledge gaps, providing the most robust evidence to date on which HRQoL measures to choose for which conditions. This study has several limitations. It was not feasible to confirm clinical details (diagnosis, severity and change in health) with treating clinicians. However, we used validated condition-specific measures to understand condition severity and symptoms. Nonetheless, our responsiveness evidence should be considered

a conservative estimate, with improved performance possible alongside a clinical intervention. This may be why only small changes are seen in the responsiveness assessments. A recent study explored this phenomenon, explaining how small effect sizes should be expected in community samples, such as ours, which do not have a dedicated intervention.²⁹ Small sample sizes limited the interpretability of test-retest reliability and responsiveness results in some condition groups and for the AQoL-6D, HUI3 and PROMIS-25. Test-retest and responsiveness are the hardest measurement attributes to assess and are where the greatest evidence gap exists.^{2 3} Consequently, the ability of this study to contribute some test-retest and responsiveness evidence to this gap is a strength. Also, the follow-up survey response rate was low in some condition groups, ranging between 28% and 50%, limiting sample sizes in test-retest and responsiveness analyses. The response rate for the follow-up survey and reasons for variation between groups have been explored in a previous study.³⁰ Furthermore, a single total score was needed for each measure to enable a head-to-head comparison. However, some measures, such as the PROMIS-25, EQ-5D-Y-3L and EQ-5D-Y-5L, may be more suited to other scoring approaches, such as domain scores or utility scores.^{14 18} The use of a single total score for the PROMIS-25, in particular, is explorative, and sensitivity analysis of the PROMIS-25 when scored as per developer instructions demonstrated weaker performance for ceiling effects and known-groups validity when compared with the exploratory total score approach. Combining proxy and child self-reports prevents us from examining differences by reporter type; however, splitting groups would have resulted in insufficient power to assess key findings, particularly responsiveness.

A previous study comparing the EQ-5D-Y-3L, PedsQL and PROMIS-25 in children with spinal muscular atrophy found the EQ-5D-Y-3L was superior to these measures.³¹ Unlike our study, this study found no evidence of known-groups validity for the PROMIS-25 and no ceiling effects for the EQ-5D-Y-3L.³¹ These differences are likely due to spinal muscular atrophy having a larger impact on HRQoL compared with conditions included in this study. Unlike the aforementioned study,³¹ we did identify ceiling effects for the EQ-5D-Y-3L and EQ-5D-Y-5L in all condition groups assessed, consistent with other literature.^{9 10 32} Ceiling effect issues were particularly pronounced in children with asthma and dental problems (between 39% and 51%),^{33–35} likely due to the intermittent nature of these conditions and the often-short recall period of the measures, a similar problem reported in adult episodic conditions.³⁶

Previously, in a broader population of 5945 Australian children both with and without health conditions from the P-MIC study, the EQ-5D-Y-5L and CHU9D were the top-performing measures.⁹ However, the current study identified the PedsQL as a top-performing measure in many condition groups. Weaker test-retest reliability was found for all measures in this study compared with the previous study.⁹ This difference is likely due to differences in the timing of the measurement of test-retest reliability; a follow-up survey at 4 weeks in this study compared with 2 days in the other study.⁹ A key finding from this study is that although some measures may perform well in a broad sample of children, they may not perform well in specific condition populations, again highlighting the importance of generating validity evidence for measures in their intended populations of use.

CONCLUSION

Using the most comprehensive dataset available, our evidence suggests that the performance of measures varies by childhood

condition. Although the PedsQL performed consistently well across the range of conditions assessed, future users should assess evidence to select a measure that best fits their intended population and use. This study suggests the PedsQL might be preferred for asthma, sleep problems, dental problems and enuresis; the PedsQL and CHU9D might be preferred for eating disorder(s), epilepsy and T1D; the PedsQL and EQ-5D-Y-5L might be preferred for ASD, and the PedsQL, EQ-5D-Y-5L and CHU9D might be preferred for recurrent abdominal pain. Importantly, small sample sizes limited the ability to interpret test-retest reliability and responsiveness results for the AQoL-6D, HUI3 and PROMIS-25 in this study; however, these measures did perform well in all other measurement attributes, warranting further investigation into the performance of these measures.

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